

Reproducing Test-Time Diffusion Alignment

DAS, Feynman–Kac Steering, Feynman–Kac Correctors, and Null-TTA

A run is not a reproduction until the metric, target, uncertainty, and evidence scope all match the paper claim.

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Evidence contract: four questions before a verdict

1. Identity

code · config · checkpoint

2. Execution

payload complete · artifacts saved

3. Numeric match

metric direction · fixed tolerance

4. Stability

all seeds · all runs · no cherry-picking

Paper	Verdict in reproduced scope	Scope anchor
DAS	PASS WITH VARIANCE	toy GMM + Kaggle portability
FK Steering	PARTIAL	one prompt, three seeds
FK Correctors	PASS WITH VARIANCE	exact Table A1
Null-TTA	PASS	SD-v1.5 Table 1, 3 seeds × 50 prompts

Evidence: final reproduction report and results/*.json

DAS: reproduce the tilted distribution, not the reward maximum

Target distribution

$$p_{\text{tar}}(x) \propto p_{\text{pre}}(x) \exp(r(x)/\alpha),$$
$$r(x_1, x_2) = -x_1^2/100 - x_2^2.$$

Full three-mode distribution.

RNG bug

NumPy seed $\neq$ Numba-jitted resampler state.

Earlier “same seed” $\neq$ same run.

canonical seed 42: 0.743

seed 43: 1.826

- ▶ Figure 1 toy GMM · seeds 42–44.
- ▶ Matched Python · Torch · NumPy · **Numba RNG.**
- ▶ Paper: EMD 0.82 · reward -0.22 .

DAS paper Figure 1; upstream commit 2f4b223; results/das_rescue_multiseed.json

DAS metric: EMD for fidelity; reward as guardrail

Primary: empirical 1-Wasserstein / EMD ↓

$$\widehat{W}_1(X, Y) = \min_{\pi \in \Pi(u_X, u_Y)} \sum_{i,j} \pi_{ij} \|x_i - y_j\|_2.$$

500 × 500 samples; 100 repeats expose estimator noise.

Secondary: mean reward

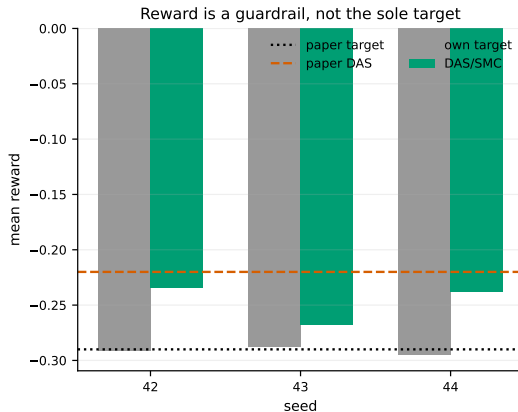
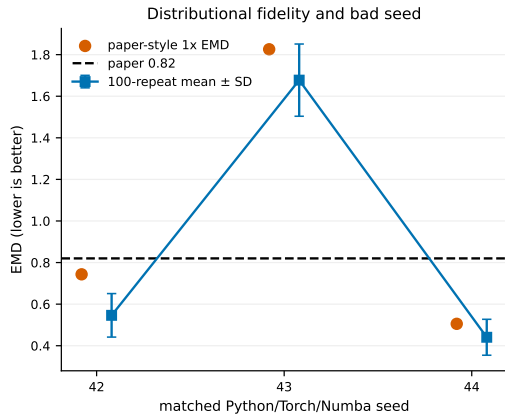
Higher reward does not guarantee correct mode mass.

Object	Reward
Target	−0.29
Paper DAS	−0.22
Repro seed 42	−0.235
Hypothetical	−0.21

Never rank −0.21 > −0.22 without first checking EMD and mode mass.

Metric contract: reproduction report Sec. DAS; DAS official GMM notebook

DAS canonical point matches, but seed 43 breaks stability



seed 42: 9.8% from paper

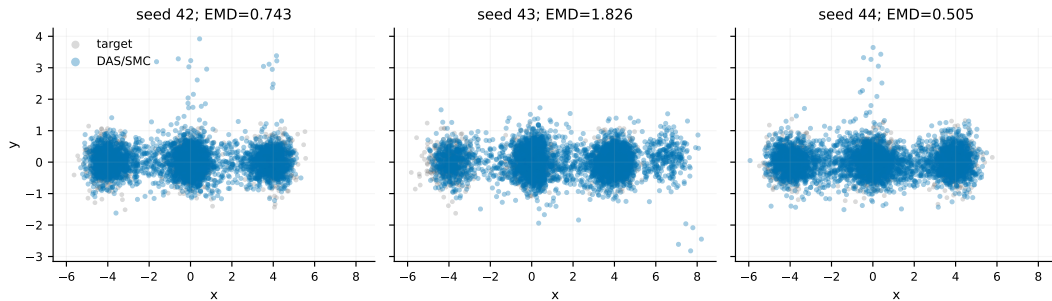
1.025 ± 0.704 across seeds

worst: 1.826

Reproduction figure from results/das_rescue_multiseed.json

All DAS seeds cover 3/3 modes, yet mass allocation still differs

DAS target samples versus SMC outputs



Seed 43:

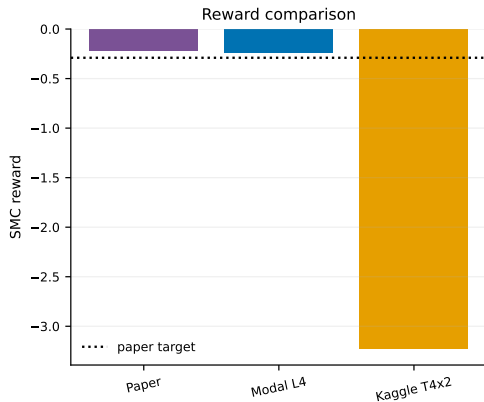
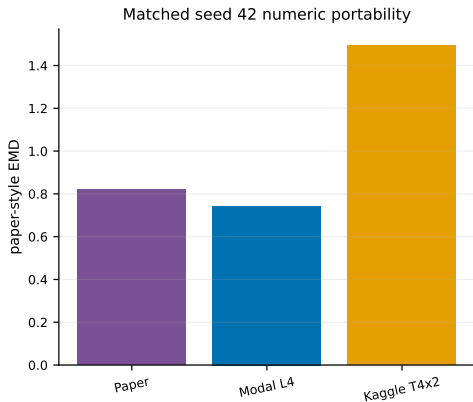
3/3 modes

mass $L_1 = 0.420$

EMD = 1.826

Saved target/SMC arrays; results/das_rescue_multiseed.json

Kaggle T4x2 executes the payload, but does not reproduce the number



Infrastructure pass: 2 Tesla T4 visible

Paper match fail: EMD 1.496 vs 0.820

Modal portability fail: 67.2% symmetric difference

T4x2 = allocation/portability test; upstream workload uses one GPU.

Kaggle r6 artifact; results/das_kaggle_t4x2_rescue.json

FK Steering: a general particle framework, reproduced at one prompt

Configuration

“brown knife + blue donut”

SD 2.1 mirror · seeds 42–44

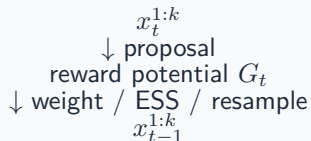
$k = 4$ · 100 steps · $\lambda = 2$

Potential rules

FK-max: best reward so far

FK-difference: reward increment

ESS → resample



Identity caveat

Upstream checkpoint: 401

Run: public mirror

FK Steering paper; upstream commit 9413005; results/fk_steering_one_prompt.json

FK Steering metric: average ImageReward across all four particles

Per method and seed

$$\bar{R}_{m,s} = \frac{1}{4} \sum_{j=1}^4 S_{\text{IR}}(x_{m,s,j}, c).$$

learned image-text score · higher is better

Aggregation

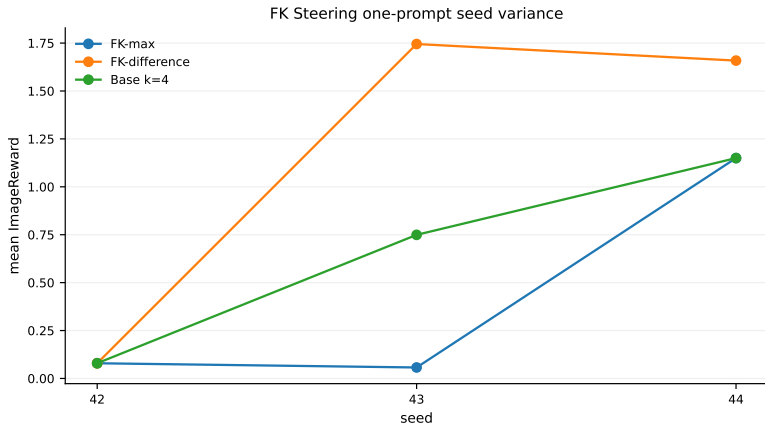
mean $\pm$ sample SD across three seeds
all particles retained

Method	mean	SD
FK-max	0.429	0.625
FK-difference	1.161	0.938
Base	0.660	0.541

Best-of-4 $\neq$ population mean

ImageReward aggregation reconstructed from the saved nine generations and final_metrics.json

FK Steering: ranking flips across three seeds



ranking flips

seed 42: all equal

seed 44: FK-max = base

Reproduction figure from results/fk_steering_one_prompt.json

FK Steering produces plausible samples; the scientific claim remains partial

FK-max, seed 43

a photo of a brown knife and a blue donut



FK-difference, seed 43

a photo of a brown knife and a blue donut



Base, seed 43

a photo of a brown knife and a blue donut



Scientific payload complete

one prompt + three seeds

not a GenEval reproduction or superiority claim

Saved remote grids; one-prompt evidence only

FK Correctors: exact Table A1 protocol

Protocol item	Reproduction
Samples per run	10,000
Integration steps	1,000
Step size dt	0.001
Stochastic runs	5
Proposals	target score, tempered noise
Correctors	none, BDC, systematic
Metric cells	$6 \times 5 = 30$

Runtime repair

Forward one existing BDC shape flag.

Hashes retained · upstream clean.

17.4 min

7.99 GiB

\$0.367

FK Correctors Table A1; upstream commit aa6f5ed; exact-run manifest

FK Correctors metrics compare different projections of error

W_1 and W_2 ↓

$$W_p = \left[\min_{\pi} \sum_{ij} \pi_{ij} \|x_i - y_j\|_2^p \right]^{1/p}.$$

2D transport · W_2 penalizes long moves.

MMD ↓

biased $\widehat{\text{MMD}}^2$ · 10 RBF kernels
bandwidth-dependent scale

TV ↓

200 × 200 histogram

$$\widehat{\text{TV}} = \frac{1}{2} \sum_b |\hat{p}_b - \hat{q}_b|$$

grid/bin-dependent

Energy- W_2 ↓

$x \mapsto E(x)$ · 1D transport
implementation returns W_2^2 , not W_2

Metric definitions audited against the official FK Correctors notebook

Energy-W2 is squared transport between energy histograms

Code semantics

`pot.emd2_1d` · squared cost · no square root

$$\begin{aligned} D_E &= \min_{\gamma} \sum_{ij} \gamma_{ij} |E(x_i) - E(y_j)|^2 \\ &= W_2^2(E_{\#}X, E_{\#}Y). \end{aligned}$$

$E_{\#}X$: distribution of scalar energy only.

$D_E = 0$, but geometry differs

$$E(x) = \|x\|_2^2,$$

$$X = \{(-1, 0), (1, 0)\},$$

$$Y = \{(0, -1), (0, 1)\}.$$

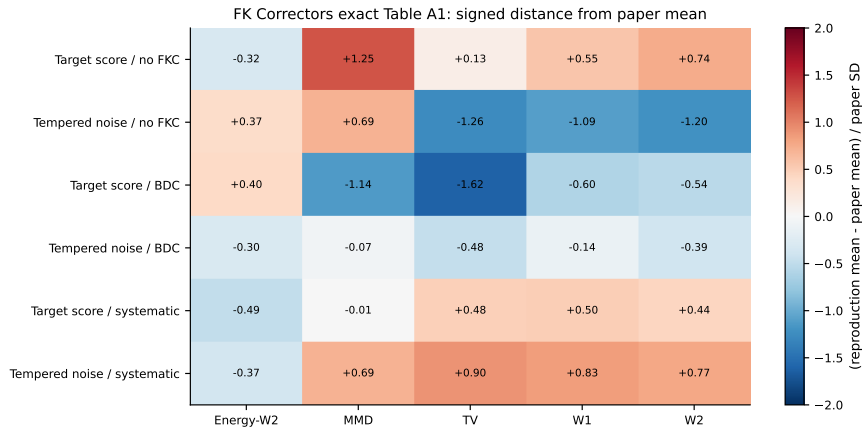
$$E_{\#}X = E_{\#}Y = \delta_1 \Rightarrow D_E = 0,$$

$$W_1(X, Y) = W_2(X, Y) = \sqrt{2} > 0.$$

- ▶ Units: energy²; $E \mapsto 2E \Rightarrow D_E \mapsto 4D_E$.
- ▶ Not a measure of 2D sample geometry.

Implementation semantics: official notebook call to `pot.emd2_1d`; example is explanatory

FK Correctors reproduces all 30 Table A1 cells within ± 2 paper SD



24/30 within ± 1 SD

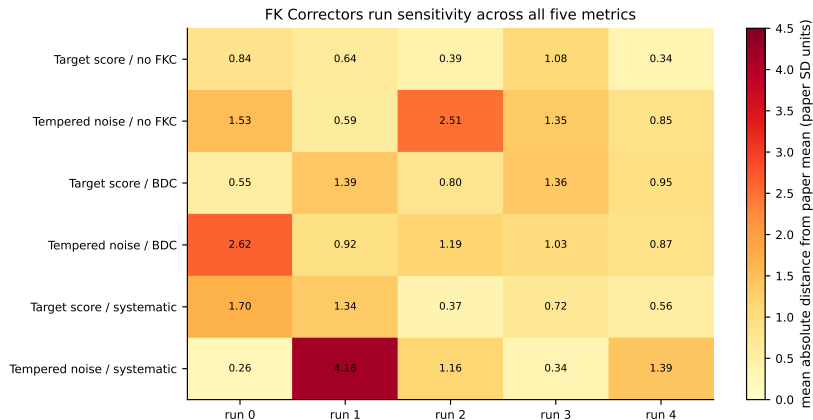
19/20 correction directions agree

$\pm$ SD is a compatibility band, not an equivalence test.

30/30 within ± 2 SD

Exact Table A1 aggregation: results/fk_correctors_table_a1_exact.json

FK Correctors: compatible means hide bad runs



BDC run 0: $W_2 = 29.925$ systematic run 1: $W_2 = 20.849$

Verdict: PASS WITH VARIANCE – exact protocol and means match, run stability does not.

Five sequential stochastic runs; upstream loop does not explicitly reseed

Null-TTA: optimize only the null-text embedding at inference

Target

Table 1 · SD-v1.5 · PickScore

$n_{\max} = 55$ · diffusion weights frozen

- ▶ 50 prompts $\times$ 3 seeds = 150 pairs.
- ▶ $K = 3$ · 100 steps · $n : 5 \rightarrow 55$.
- ▶ Same prompt/seed: baseline vs optimized.

PickScore $\uparrow$

cosine(normalized image, text)

no softmax · not a probability

Gate

symmetric difference $\leq 10\%$

95% CI(Δ) entirely > 0

Null-TTA official code commit 337bf73; results/null_tta_table1.json

Null-TTA: three secondary alignment proxies

HPS v2 ↑

held-out text–image
preference proxy

Aesthetic ↑

image-only CLIP feature →
MLP

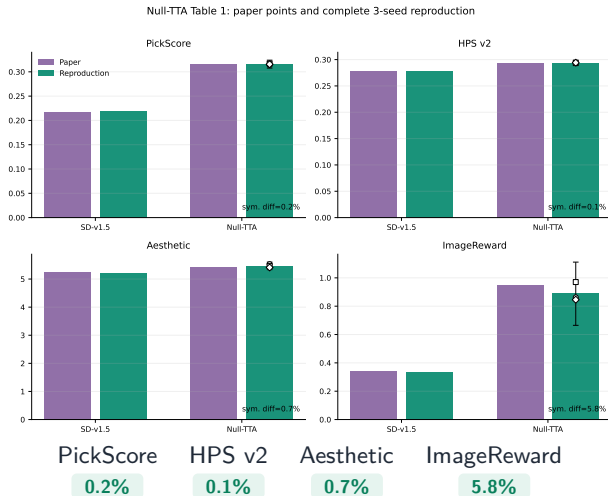
ImageReward ↑

BLIP image–text reward ·
unbounded

Proxy agreement supports transfer; **none replaces human evaluation.**

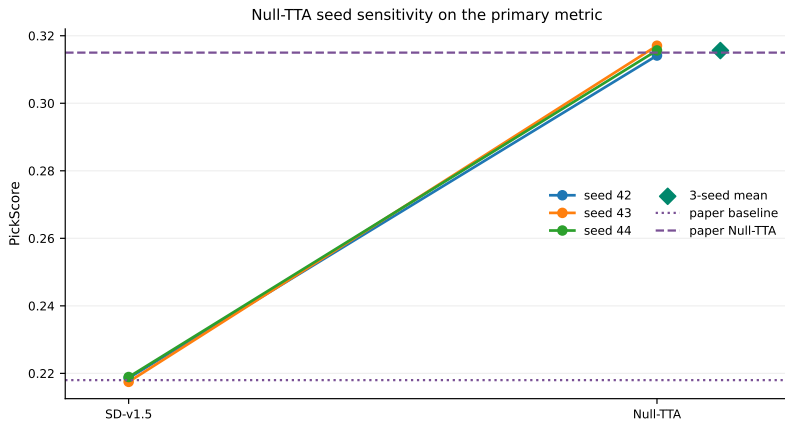
Scorer implementations and checkpoints audited in the reproduction report

Null-TTA matches all four Table 1 point estimates



Paper and reproduction rows from results/null_tta_table1.json

Null-TTA: stable PickScore gain across seeds



repro

0.316

paper

0.315

paired gain

0.097

95% CI

[0.089, 0.106]

50 prompts per seed; hierarchical bootstrap resamples seed then prompt, 10,000 repeats

Null-TTA: qualitative pairs support the scoped result

Bus: baseline, PickScore 0.213



Bus: optimized, 0.313



Bikes: baseline, 0.226



Bikes: optimized, 0.330



PASS for SD-v1.5 Table 1

not evidence for SDXL, other rewards, or universal human preference

Seed 44 official PNG artifacts; prompts 0–1; PickScore from saved CSV

What the reproduction supports – and what it does not

Paper	Verdict	Evidence boundary
DAS	PASS + VAR	seed 42 matches · seed variance · Kaggle fail
FK Steering	PARTIAL	one prompt · three seeds · model mirror
FK Correctors	PASS + VAR	30/30 compatible · bad sequential runs
Null-TTA	PASS	150 pairs · four metrics · SD-v1.5 only

Narrow · stable · traceable claims beat the best-looking score.

All verdicts are scoped to executed artifacts; no benchmark-wide extrapolation